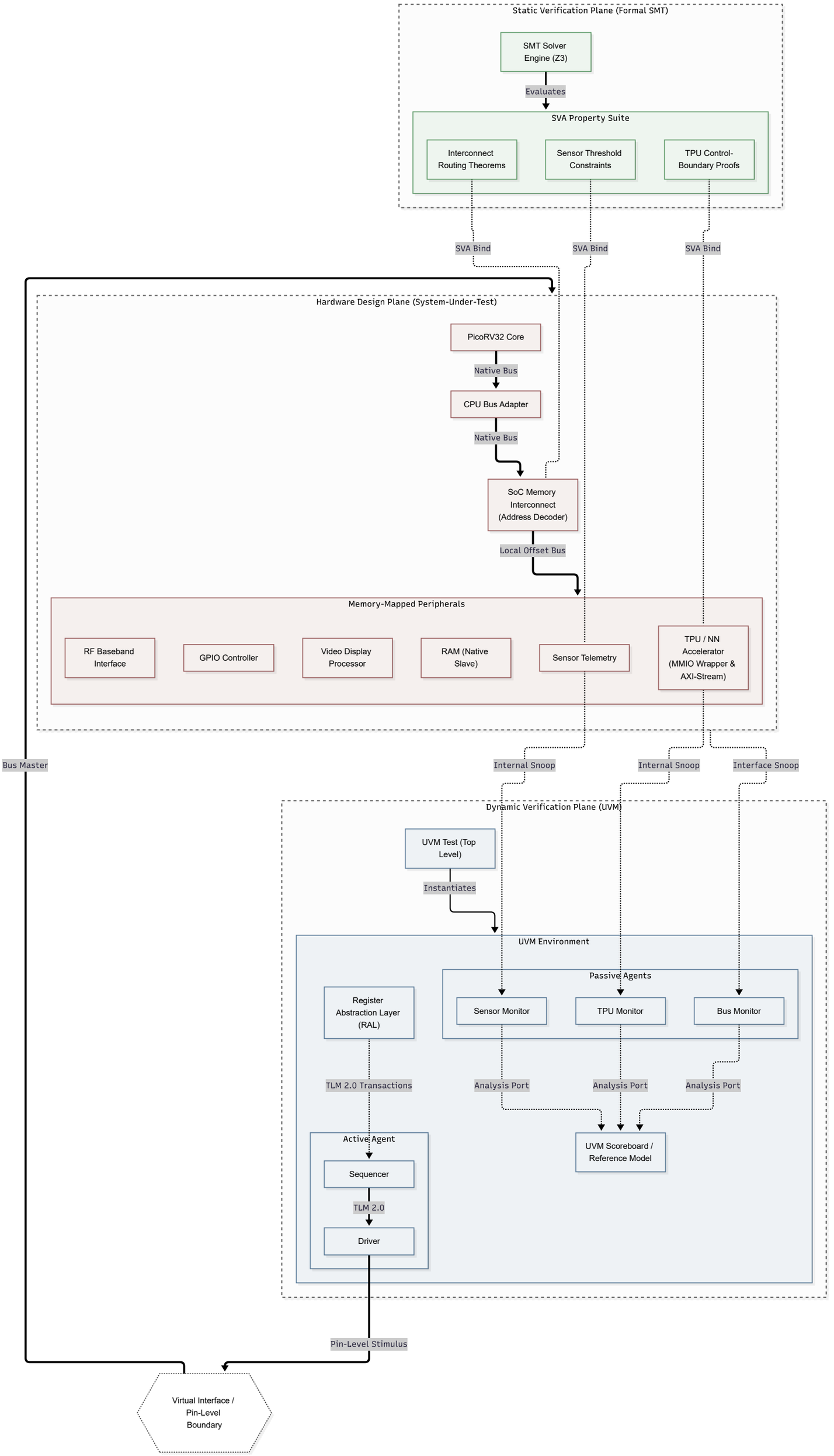

Static Verification Plane (Formal SMT)
SMT Solver Engine (Z3)
Evaluates
SVA Property Suite
Interconnect Routing Theorems
Sensor Threshold Constraints
TPU Control-Boundary Proofs
SVA Bind
SVA Bind
SVA Bind
Hardware Design Plane (System-Under-Test)
PicoRV32 Core
Native Bus
CPU Bus Adapter
Native Bus
SoC Memory Interconnect (Address Decoder)
Local Offset Bus
Memory-Mapped Peripherals
RF Baseband Interface
GPIO Controller
Video Display Processor
RAM (Native Slave)
Sensor Telemetry
TPU / NN Accelerator (MMIO Wrapper & AXI-Stream)
Bus Master
Internal Snoop
Internal Snoop
Interface Snoop
Dynamic Verification Plane (UVM)
UVM Test (Top Level)
Instantiates
UVM Environment
Passive Agents
Register Abstraction Layer (RAL)
Sensor Monitor
TPU Monitor
Bus Monitor
TLM 2.0 Transactions
Analysis Port
Analysis Port
Analysis Port
Active Agent
UVM Scoreboard / Reference Model
Sequencer
TLM 2.0
Driver
Pin-Level Stimulus
Virtual Interface / Pin-Level Boundary